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DBMS Lab 11

Problem 1: WorkSphere – Enterprise Collaboration Platform

Task 1: Setup
1.Create the database and all collections.
2. Insert at least 10 records in each collection ensuring variation in departments, roles, and activities.

Task 2: Query Implementation

1. Retrieve all active users from the IT department who joined after the year 2022. Display relevant user details.

```
> db.users.find({department: "IT",joiningDate: { $gte: new Date("2023-01-01")}})
< {
    _id: ObjectId('69d62465cf532a1b719965d1'),
    userId: 'U005',
    name: 'Name5',
    department: 'IT',
    role: 'Security Analyst',
    joiningDate: 2023-09-12T00:00:00.000Z,
    isActive: true
  }
```

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2. List all projects having a budget greater than 100000 and currently marked as “Ongoing”. Sort results by budget in descending order.

```
{
  _id: ObjectId('69d6254ccf532a1b719965dc'),
  projectId: 'P006',
  projectName: 'Project6',
  ownerId: 'U006',
  budget: 300000,
  status: 'ongoing',
  tags: [
    'cloud',
    'migration'
  ],
  deadline: 2026-08-01T00:00:00.000Z
}
{
  _id: ObjectId('69d6254ccf532a1b719965da'),
  projectId: 'P004',
  projectName: 'Project4',
  ownerId: 'U004',
  budget: 200000,
  status: 'ongoing',
  tags: [
    'security',
    'audit'
  ],
  deadline: 2026-07-10T00:00:00.000Z
}
```

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```
{
  _id: ObjectId('69d6254ccf532a1b719965de'),
  projectId: 'P008',
  projectName: 'Project8',
  ownerId: 'U008',
  budget: 150000,
  status: 'ongoing',
  tags: [
    'mobile',
    'app'
  ],
  deadline: 2026-09-15T00:00:00.000Z
}
{
  _id: ObjectId('69d6254ccf532a1b719965d8'),
  projectId: 'P002',
  projectName: 'Project2',
  ownerId: 'U002',
  budget: 120000,
  status: 'ongoing',
  tags: [
    'engineering',
    'backend'
  ],
  deadline: 2026-06-30T00:00:00.000Z
}
```

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```
{
  _id: ObjectId('69d6254ccf532a1b719965e0'),
  projectId: 'P010',
  projectName: 'Project10',
  ownerId: 'U010',
  budget: 110000,
  status: 'ongoing',
  tags: [
    'ai',
    'ml'
  ],
  deadline: 2026-10-01T00:00:00.000Z
}
```

3. Identify all activity records where time spent exceeds 5 hours. Sort them from highest to lowest.

```
> db.activities.find({timeSpentHours : {$gt : 5}}).sort({timeSpentHours:-1});
< {
  _id: ObjectId('69d626e43a95afe3a58f6cd0'),
  userId: 'U009',
  projectId: 'P009',
  action: 'Feature Development',
  timeSpentHours: 6,
  timestamp: 2026-04-09T07:15:00.000Z
}
```

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4. Compute the total number of hours contributed by each user across all projects using aggregation.

```
    totalHours: 6,
    userId: 'U009',
    name: 'Name9',
    department: 'Finance',
    role: 'Financial Analyst'
  }
  {
    totalHours: 5,
    userId: 'U006',
    name: 'Name6',
    department: 'Marketing',
    role: 'Marketing Specialist'
  }
  {
    totalHours: 4,
    userId: 'U002',
    name: 'Name2',
    department: 'HR',
    role: 'HR Manager'
  }
  {
    totalHours: 3.5,
    userId: 'U008',
    name: 'Name8',
    department: 'Design',
    role: 'UI/UX Designer'
  }
```

```
  totalHours: 3,
  userId: 'U003',
  name: 'Name3',
  department: 'Finance',
  role: 'Accountant'
}
{
  totalHours: 2.5,
  userId: 'U001',
  name: 'Name1',
  department: 'Engineering',
  role: 'Software Developer'
}
{
  totalHours: 2,
  userId: 'U005',
  name: 'Name5',
  department: 'IT',
  role: 'Security Analyst'
}
{
  totalHours: 2,
  userId: 'U010',
  name: 'Name10',
  department: 'Engineering',
  role: 'Team Lead'
}
{
```

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```
{
  totalHours: 1.5,
  userId: 'U004',
  name: 'Name4',
  department: 'Engineering',
  role: 'DevOps Engineer'
}
{
  totalHours: 1,
  userId: 'U007',
  name: 'Name7',
  department: 'Engineering',
  role: 'QA Engineer'
}
```

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5. Determine the top 5 most active users based on the number of activity records.

```
> db.activities.aggregate([
  {$group:{_id:"$userId",activityCount:{$sum:1}}},
  {$sort:{activityCount:-1}},
  {$limit:5}])
< {
    _id: 'U002',
    activityCount: 1
  }
  {
    _id: 'U006',
    activityCount: 1
  }
  {
    _id: 'U003',
    activityCount: 1
  }
  {
    _id: 'U004',
    activityCount: 1
  }
  {
    _id: 'U001',
    activityCount: 1
  }
```

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6. Calculate the average time spent per project.

```
> db.activities.aggregate([
  {$group:{_id:"$projectId",avgTime:{$avg:"$timeSpentHours"}}}
  ])
< {
    _id: 'P004',
    avgTime: 1.5
  }
  {
    _id: 'P003',
    avgTime: 3
  }
  {
    _id: 'P001',
    avgTime: 2.5
  }
  {
    _id: 'P002',
    avgTime: 4
  }
  {
    _id: 'P005',
    avgTime: 2
  }
  {
    _id: 'P007',
    avgTime: 1
```

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```
{
  _id: 'P007',
  avgTime: 1
}
{
  _id: 'P006',
  avgTime: 5
}
{
  _id: 'P009',
  avgTime: 6
}
{
  _id: 'P010',
  avgTime: 2
}
{
  _id: 'P008',
  avgTime: 3.5
}
```

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7. Using lookup, display combined information showing user name, project name, and action performed.

```
> db.activities.aggregate([
  {$lookup:{from:"users",localField:"userId",foreignField:"userId",as:"userInfo"}},
  {$lookup:{from:"projects",localField:"projectId",foreignField:"projectId",as:"projectInfo"}},
  {$unwind:"$userInfo"},
  {$unwind:"$projectInfo"},
  {$project:{_id:0,name:"$userInfo.name",projectName:"$projectInfo.projectName",action:1}}])
< {
    action: 'Code Review',
    name: 'Name1',
    projectName: 'Project1'
  }
  {
    action: 'Design Mockups',
    name: 'Name2',
    projectName: 'Project2'
  }
  {
    action: 'Database Optimization',
    name: 'Name3',
    projectName: 'Project3'
  }
  {
    action: 'Testing',
    name: 'Name4',
    projectName: 'Project4'
  }
```

```
  action: 'Documentation',
  name: 'Name5',
  projectName: 'Project5'
}
{
  action: 'Deployment',
  name: 'Name6',
  projectName: 'Project6'
}
{
  action: 'Client Meeting',
  name: 'Name7',
  projectName: 'Project7'
}
{
  action: 'Bug Fixing',
  name: 'Name8',
  projectName: 'Project8'
}
{
  action: 'Feature Development',
  name: 'Name9',
  projectName: 'Project9'
}
{
  action: 'Code Merge',
  name: 'Name10',
  projectName: 'Project10'
```

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8. Identify projects whose deadlines fall within the next 30 days.

```
> db.projects.aggregate([{$match:{deadline:{$gte:new Date(),$lte:new Date(new Date().getTime()+30*24*60*60*1000)}}}])
< {
    _id: ObjectId('69d6254ccf532a1b719965d9'),
    projectId: 'P003',
    projectName: 'Project3',
    ownerId: 'U003',
    budget: 95000,
    status: 'ongoing',
    tags: [
      'design',
      'ui'
    ],
    deadline: 2026-04-25T00:00:00.000Z
  }
  {
    _id: ObjectId('69d6254ccf532a1b719965db'),
    projectId: 'P005',
    projectName: 'Project5',
    ownerId: 'U005',
    budget: 75000,
    status: 'completed',
    tags: [
      'marketing',
      'campaign'
    ],
    deadline: 2026-04-20T00:00:00.000Z
  }
```

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```
{
  _id: ObjectId('69d6254ccf532a1b719965dd'),
  projectId: 'P007',
  projectName: 'Project7',
  ownerId: 'U007',
  budget: 40000,
  status: 'pending',
  tags: [
    'training',
    'hr'
  ],
  deadline: 2026-05-05T00:00:00.000Z
}
{
  _id: ObjectId('69d6254ccf532a1b719965df'),
  projectId: 'P009',
  projectName: 'Project9',
  ownerId: 'U009',
  budget: 60000,
  status: 'completed',
  tags: [
    'analytics',
    'dashboard'
  ],
  deadline: 2026-04-28T00:00:00.000Z
}
```

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9. Determine the most frequently used tag across all projects.

```
> db.projects.aggregate([
  {$unwind:"$tags"},
  {$group:{_id:"$tags",count:{$sum:1}}},
  {$sort:{count:-1}},
  {$limit:1}])
< {
    _id: 'cloud',
    count: 1
  }
```

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10. Compute total contribution (in hours) grouped by department.

```
> db.activities.aggregate([
  {$lookup:{from:"users",localField:"userId",foreignField:"userId",as:"userInfo"}},
  {$unwind:"$userInfo"},
  {$group:{_id:"$userInfo.department",totalHours:{$sum:"$timeSpentHours"}}}])
< {
    _id: 'IT',
    totalHours: 2
  }
  {
    _id: 'HR',
    totalHours: 4
  }
  {
    _id: 'Engineering',
    totalHours: 7
  }
  {
    _id: 'Finance',
    totalHours: 9
  }
  {
    _id: 'Marketing',
    totalHours: 5
  }
  {
    _id: 'Design',
    totalHours: 3.5
  }
```

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Problem 2: ShopStream – E-commerce Analytics

Task 1: Setup 1. Create the database and insert at least 10 records per collection.

Task 2: Query Implementation -

1. Retrieve all customers from Delhi having Gold membership and display their details.

```
> db.customers.find({city:"Delhi",membership:"Gold"})
< {
    _id: ObjectId('69d63056985e5dd464907455'),
    customerId: 'C001',
    name: 'Amit Sharma',
    city: 'Delhi',
    membership: 'Gold',
    joinDate: 2023-01-15T00:00:00.000Z
  }
```

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2. List products whose price exceeds 50000, sorted in descending order.

```
> db.products.find({price:{$gt:50000}}).sort({price:-1})
< {
    _id: ObjectId('69d630af985e5dd46490745f'),
    productId: 'P001',
    name: 'Laptop Pro X',
    category: 'Electronics',
    price: 75000,
    stock: 20
  }
  {
    _id: ObjectId('69d630af985e5dd464907460'),
    productId: 'P002',
    name: 'Smartphone Ultra',
    category: 'Electronics',
    price: 65000,
    stock: 35
  }
  {
    _id: ObjectId('69d630af985e5dd464907467'),
    productId: 'P009',
    name: 'Refrigerator XL',
    category: 'Appliances',
    price: 60000,
    stock: 12
  }
```

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```
{
  _id: ObjectId('69d630af985e5dd464907462'),
  productId: 'P004',
  name: 'Electric Scooter',
  category: 'Automobile',
  price: 55000,
  stock: 15
}
{
  _id: ObjectId('69d630af985e5dd464907465'),
  productId: 'P007',
  name: 'Smart TV 55inch',
  category: 'Electronics',
  price: 52000,
  stock: 18
}
```

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3. Find all orders placed within the last 6 months.

```
> db.orders.find({orderDate: {
  $gte: new Date(new Date().setMonth(new Date().getMonth()-6))}
< {
    _id: ObjectId('69d63117985e5dd46490746b'),
    orderId: 'O003',
    customerId: 'C003',
    productId: 'P002',
    quantity: 3,
    totalAmount: 195000,
    orderDate: 2026-01-10T00:00:00.000Z
  }
  {
    _id: ObjectId('69d63117985e5dd46490746e'),
    orderId: 'O006',
    customerId: 'C006',
    productId: 'P007',
    quantity: 2,
    totalAmount: 104000,
    orderDate: 2026-02-14T00:00:00.000Z
  }
  {
    _id: ObjectId('69d63117985e5dd464907470'),
    orderId: 'O008',
    customerId: 'C008',
    productId: 'P009',
    quantity: 1,
    totalAmount: 60000,
    orderDate: 2025-12-01T00:00:00.000Z
```

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```
{
  _id: ObjectId('69d63117985e5dd464907471'),
  orderId: 'O009',
  customerId: 'C009',
  productId: 'P010',
  quantity: 2,
  totalAmount: 70000,
  orderDate: 2026-03-22T00:00:00.000Z
}
```

4. Compute total revenue generated for each product.

```
> db.orders.aggregate([
    {$group:{_id:"$productId",totalRevenue:{$sum:"$totalAmount"}}},
    {$sort:{totalRevenue:-1}}
  ])
< {
    _id: 'P002',
    totalRevenue: 195000
  }
  {
    _id: 'P001',
    totalRevenue: 150000
  }
  {
    _id: 'P007',
    totalRevenue: 104000
  }
  {
    _id: 'P006',
    totalRevenue: 90000
  }
  {
    _id: 'P005',
    totalRevenue: 80000
  }
  {
    _id: 'P010',
    totalRevenue: 70000
  }
```

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```
{
  _id: 'P010',
  totalRevenue: 70000
}
{
  _id: 'P009',
  totalRevenue: 60000
}
{
  _id: 'P004',
  totalRevenue: 55000
}
{
  _id: 'P008',
  totalRevenue: 28000
}
{
  _id: 'P003',
  totalRevenue: 12000
}
```

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5. Identify the top 3 selling products based on total quantity sold.

```
> db.orders.aggregate([
  {$group:{_id:"$productId",totalQuantity:{$sum:"$quantity"}}},
  {$sort:{totalQuantity:-1}},
  {$limit:3}
  ])
< {
    _id: 'P006',
    totalQuantity: 3
  }
  {
    _id: 'P002',
    totalQuantity: 3
  }
  {
    _id: 'P007',
    totalQuantity: 2
  }
```

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6. Calculate average order value per customer.

```
> db.orders.aggregate([
  {$group:{_id:"$customerId",avgOrderValue:{$avg:"$totalAmount"}}
  ])
< {
    _id: 'C008',
    avgOrderValue: 60000
  }
  {
    _id: 'C007',
    avgOrderValue: 28000
  }
  {
    _id: 'C003',
    avgOrderValue: 195000
  }
  {
    _id: 'C001',
    avgOrderValue: 150000
  }
  {
    _id: 'C002',
    avgOrderValue: 12000
  }
  {
    _id: 'C005',
    avgOrderValue: 55000
  }
```

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```
{
  _id: 'C006',
  avgOrderValue: 104000
}
{
  _id: 'C004',
  avgOrderValue: 80000
}
{
  _id: 'C010',
  avgOrderValue: 90000
}
{
  _id: 'C009',
  avgOrderValue: 70000
}
```

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7. Using lookup, display customer name, product name, and order amount together.

```
> db.orders.aggregate([
  {$lookup:{from:"customers",localField:"customerId",foreignField:"customerId",as:"customerInfo"}},
  {$lookup:{from:"products",localField:"productId",foreignField:"productId",as:"productInfo"}},
  {$unwind:"$customerInfo"},
  {$unwind:"$productInfo"},
  {$project:{_id:0,name:"customerInfo.name",productName:"productInfo.name",orderAmount:"$totalAmount"}}
  ]);
< {
    name: 'customerInfo.name',
    productName: 'productInfo.name',
    orderAmount: 150000
  }
  {
    name: 'customerInfo.name',
    productName: 'productInfo.name',
    orderAmount: 12000
  }
  {
    name: 'customerInfo.name',
    productName: 'productInfo.name',
    orderAmount: 195000
  }
  {
    name: 'customerInfo.name',
    productName: 'productInfo.name',
    orderAmount: 80000
```

```
{
  name: 'customerInfo.name',
  productName: 'productInfo.name',
  orderAmount: 55000
}
{
  name: 'customerInfo.name',
  productName: 'productInfo.name',
  orderAmount: 104000
}
{
  name: 'customerInfo.name',
  productName: 'productInfo.name',
  orderAmount: 28000
}
{
  name: 'customerInfo.name',
  productName: 'productInfo.name',
  orderAmount: 60000
}
{
  name: 'customerInfo.name',
  productName: 'productInfo.name',
  orderAmount: 70000
}
{
  name: 'customerInfo.name',
  productName: 'productInfo.name',
  orderAmount: 90000
```

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8. Compute monthly sales trend for the year 2024 using aggregation.

```
> db.orders.aggregate([
  {$match:{orderDate:{$gte:new Date("2024-01-01"),$lte:new Date("2024-12-31")}}},
  {$group:{_id:{month:{$month:"$orderDate"}},monthlySales:{$sum:"$totalAmount"}}},
  {$sort:{"_id.month":1}}
  ])
< {
    _id: {
      month: 3
    },
    monthlySales: 150000
  }
  {
    _id: {
      month: 7
    },
    monthlySales: 80000
  }
  {
    _id: {
      month: 8
    },
    monthlySales: 90000
  }
  {
    _id: {
      month: 11
    },
    monthlySales: 28000
  }
```

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9. Identify customers who have placed more than 5 orders.

```
> db.orders.aggregate([
  {$group:{_id:"$customerId",orderCount:{$sum:1}}},
  {$match:{orderCount:{$gte:5}}}
  ]);
< {
    _id: 'C001',
    orderCount: 6
  }
```

10. Compute total revenue grouped by product category.

```
> db.orders.aggregate([
  {$lookup:{from:"products",localField:"productId",foreignField:"productId",as:"productInfo"}},
  {$unwind:"$productInfo"},
  {$group:{_id:"$productInfo.category",totalRevenue:{$sum:"$totalAmount"}}}
  ])
< {
    _id: 'Furniture',
    totalRevenue: 40000
  }
  {
    _id: 'Electronics',
    totalRevenue: 539000
  }
  {
    _id: 'Appliances',
    totalRevenue: 360000
  }
  {
    _id: 'Automobile',
    totalRevenue: 55000
  }
```